PULLBACK OF A QUASI-COHERENT MODULE ALONG A MORPHISM OF SIMPLICIAL SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The label lemma-pullback-cartesian-module names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The counted proof is the author's complete argument at corollary depth; the flagged 'some details omitted' sits on the description step that the registered lemma-describe-functoriality card carries in full.

Allowed standard prerequisites (editorial).

- Functoriality of simplicial pullback. Same chapter, \label{lemma-describe-functoriality} (registered card). Line 6542: gives (f^*F)_n = f_n^*F_n.
- Characterization of quasi-coherent modules. Same chapter, \label{lemma-quasi-coherent-sheaf}. Lines 6541 and 6549: quasi-coherent iff cartesian with quasi-coherent terms.

Source reference. Original lemma 0D7M: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 2817–2817. Prerequisite pointer invoked at proof line(s) 6541; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 09VP: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 414–414. Prerequisite pointer invoked at proof line(s) 6542; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $f : V \to U$ be a morphism of simplicial schemes. Given a quasi-coherent module $\mathcal{F}$ on U_{Zar} the pullback $f^*\mathcal{F}$ is a quasi-coherent module on V_{Zar}.*

Proof. Recall that $\mathcal{F}$ is cartesian with $\mathcal{F}_n$ quasi-coherent, see Lemma 0D7M. By Lemma 09VP we see that $(f^*\mathcal{F})_n = f_n^*\mathcal{F}_n$ (some details omitted). Hence $(f^*\mathcal{F})_n$ is quasi-coherent. The same fact and the cartesian property for $\mathcal{F}$ imply the cartesian property for $f^*\mathcal{F}$. Thus $\mathcal{F}$ is quasi-coherent by Lemma 0D7M again. □

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